

soyDockDB: a Database for Comparing Docking of Natural Compounds to Chemical Fungicides

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Abstract:

Soybean is a valuable crop used for human consumption, livestock feed, and industrial purposes. Several fungal species act as soybean pathogens, significantly reducing crop yield and sustainability and necessitating fungicide use. Many existing synthetic fungicides, however, pose environmental and health risks, and their widespread and exclusive use may lead to fungi species developing resistances, decreasing their effectiveness. Natural compounds make promising candidates for alternative fungicides with decreased environmental and health risks, and their combined use (with each other or existing fungicides) lowers the risk of resistances emerging. To aide in finding these natural alternatives, we propose soyDockDB, a database for comparing 87,510 natural compounds side-by-side against existing synthetic fungicides across seven fungal proteins. Users can evaluate binding affinities (strength of interaction) and stability (duration of interaction) metrics between a fungal protein and a candidate binding molecule (ligand) to compare simulated effectiveness relative to existing fungicides. Upon completion, soyDockDB will be publicly available for use at <https://kaabil.net/soyDockDB/>. The web application driving the database is designed for high portability, and its source code is freely available on GitHub, expediting the development of similar protein-ligand databases.

Keywords: Soybean, fungicides, plant-based compounds, protein-ligand docking database, soyDockDB